

Daniele Ielmini

professor

基本信息

姓名: Daniele Ielmini 所属院校: 米兰理工大学
地区: Italy 年龄范围: 50岁以上

学科领域

材料物理 集成电路设计与集成系统 人工智能 电子信息材料 微电子科学与工程
电子与计算机工程

研究领域

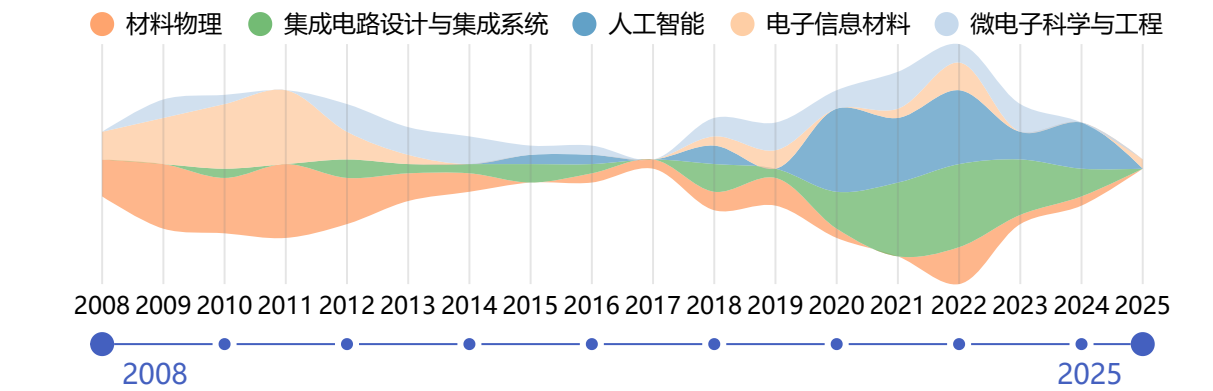
存储器设计与架构 深度学习与神经网络 半导体材料与器件 半导体与光电子材料
微电子器件与制造 纳米电子与先进工艺 纳米物理与纳米器件 嵌入式系统与硬件设计
半导体器件物理与工艺

个人简介

Daniele Ielmini is a full professor at Politecnico di Milano. He received his Laurea (cum laude) and Ph.D. in Nuclear Engineering from the same institution in 1995 and 2000. He had visiting positions at Stanford University (2006) and the University of Illinois at Urbana - Champaign (2010). His research focuses on electronics, especially electron devices, including non - volatile memories. He co - edited a book in 2016, authored/co - authored 9 book chapters, 4 patents, and over 250 papers. He serves on multiple conference sub - committees, is an Associate Editor of two journals, and a Senior Member of the IEEE. He has won several awards and was named a Highly Cited Researcher in 2015. His office is

in Building 22, Floor 4, Office 008, and his phone number is 6120. Personal page: <https://ielmini.faculty.polimi.it/>.

近年发表的论文



Optoelectronic memristor based on a-C:Te film for muti-mode reservoir computing
JOURNAL OF SEMICONDUCTORS
Tian, Qiaoling; Xun, Kuo; Li, Zhuangzhuang; Zhao, Xiaoning; Lin, Ya; Tao, Ye; Wang, Zhongqiang; Ielmini, Daniele; Xu, Haiyang; Liu, Yichun
被引用次数: 1 2025

Emerging Materials and Computing Paradigms for Temporal Signal Analysis
ADVANCED MATERIALS
Zhang, Teng; Wozniak, Stanislaw; Syed, Ghazi Sarwat; Mannocci, Piergiulio; Farronato, Matteo; Ielmini, Daniele; Sebastian, Abu; Yang, Yuchao
被引用次数: 0 2025

Seizure detection via reservoir computing in MoS2-based charge trap memory devices
SCIENCE ADVANCES
Farronato, Matteo; Mannocci, Piergiulio; Milozzi, Alessandro; Compagnoni, Christian Monzio; Barcellona, Alessandro; Arena, Andrea; Crepaldi, Marco; Panuccio, Gabriella; Ielmini, Daniele
被引用次数: 0 2025

Binary-Stochasticity-Enabled Highly Efficient Neuromorphic Deep Learning Achieves Better-than-Software Accuracy
ADVANCED INTELLIGENT SYSTEMS
Li, Yang; Wang, Wei; Wang, Ming; Dou, Chunmeng; Ma, Zhengyu; Zhou, Huihui; Zhang, Peng; Lepri, Nicola; Zhang, Xumeng; Luo, Qing; Xu, Xiaoxin; Yang, Guanhua; Zhang, Feng; Li, Ling; Ielmini, Daniele; Liu, Ming
被引用次数: 0 2024

Hemispherical Retina Emulated by Plasmonic Optoelectronic Memristors with All-Optical Modulation for Neuromorphic Stereo Vision
ADVANCED SCIENCE

Shan, Xuanyu; Wang, Zhongqiang; Xie, Jun; Han, Jiaqi; Tao, Ye; Lin, Ya; Zhao, Xiaoning; Ielmini, Daniele; Liu, Yichun; Xu, Haiyang

被引用次数: 5

2024

Memristive Materials, Devices, and Systems (MEMRISYS 2023)

ADVANCED ELECTRONIC MATERIALS

Ricciardi, Carlo; Ielmini, Daniele; Corinto, Fernando; Spiga, Sabina

被引用次数: 0

2024

Volatile and Nonvolatile Dual-Mode Switching Operations in an Ag-Ag₂S Core-Shell Nanoparticle Atomic Switch Network

ADVANCED ELECTRONIC MATERIALS

Srikimkaew, Oradee; Ricci, Saverio; Porzani, Matteo; Dang, Thien Tan; Nakaoka, Yusuke; Usami, Yuki; Ielmini, Daniele; Tanaka, Hirofumi

被引用次数: 1

2024

Roadmap to neuromorphic computing with emerging technologies

APL MATERIALS

Mehonic, Adnan; Ielmini, Daniele; Roy, Kaushik; Mutlu, Onur; Kvatsinsky, Shahar; Serrano-Gotarredona, Teresa; Linares-Barranco, Bernabe; Spiga, Sabina; Savel'ev, Sergey; Balanov, Alexander G.; Chawla, Nitin; Desoli, Giuseppe; Malavena, Gerardo; Monzio Compagnoni, Christian; Wang, Zhongrui; Yang, J. Joshua; Sarwat, Syed Ghazi; Sebastian, Abu; Mikolajick, Thomas; Slesazeck, Stefan; Noheda, Beatriz; Dieny, Bernard; Hou, Tuo-Hung (Alex); Varri, Akhil; Brueckerhoff-Plueckelmann, Frank; Pernice, Wolfram; Zhang, Xixiang; Pazos, Sebastian; Lanza, Mario; Wiefels, Stefan; Dittmann, Regina; Ng, Wing H.; Buckwell, Mark; Cox, Horatio R. J.; Mannion, Daniel J.; Kenyon, Anthony J.; Lu, Yingming; Yang, Yuchao; Querlioz, Damien; Hutin, Louis; Vianello, Elisa; Chowdhury, Sayeed Shafayet; Mannocci, Piergiulio; Cai, Yimao; Sun, Zhong; Pedretti, Giacomo; Strachan, John Paul; Strukov, Dmitri; Le Gallo, Manuel; Ambrogio, Stefano; Valov, Ilia; Waser, Rainer

被引用次数: 0

2024

In-Memory Principal Component Analysis by Analogue Closed-Loop Eigendecomposition

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Mannocci, Piergiulio; Giannone, Elisabetta; Ielmini, Daniele

被引用次数: 0

2024

Memristive tonotopic mapping with volatile resistive switching memory devices

NATURE COMMUNICATIONS

Milozzi, Alessandro; Ricci, Saverio; Ielmini, Daniele

被引用次数: 5

2024

Hardware implementation of memristor-based artificial neural networks

NATURE COMMUNICATIONS

Aguirre, Fernando; Sebastian, Abu; Le Gallo, Manuel; Song, Wenhao; Wang, Tong; Yang, J. Joshua; Lu, Wei; Chang, Meng-Fan; Ielmini, Daniele; Yang, Yuchao; Mehonic, Adnan; Kenyon, Anthony; Villena, Marco A.; Roldan, Juan B.; Wu, Yuting; Hsu, Hung-Hsi; Raghavan, Nagarajan; Sune, Jordi; Miranda, Enrique; Eltawil, Ahmed; Setti, Gianluca; Smagulova, Kamilya; Salama, Khaled N.; Krestinskaya, Olga; Yan, Xiaobing; Ang, Kah-Wee; Jain, Samarth; Li, Sifan; Alharbi, Osamah; Pazos, Sebastian; Lanza, Mario

被引用次数: 35

2024

Digital In-Memory Computing to Accelerate Deep Learning Inference on the Edge

2024 IEEE INTERNATIONAL PARALLEL AND DISTRIBUTED PROCESSING SYMPOSIUM WORKSHOPS, IPDPSW 2024

Perri, Stefania; Zambelli, Cristian; Ielmini, Daniele; Silvano, Cristina

被引用次数: 0

2024

Hybrid 2D-CMOS microchips for memristive applications

NATURE

Zhu, Kaichen; Pazos, Sebastian; Aguirre, Fernando; Shen, Yaqing; Yuan, Yue; Zheng, Wenwen; Alharbi, Osamah; Villena, Marco A.; Fang, Bin; Li, Xinyi; Milozzi, Alessandro; Farronato, Matteo; Munoz-Rojo, Miguel; Wang, Tao; Li, Ren; Fariborzi, Hossein; Roldan, Juan B.; Benstetter, Guenther; Zhang, Xixiang; Alshareef, Husam N.; Grasser, Tibor; Wu, Huaqiang; Ielmini, Daniele; Lanza, Mario

被引用次数: 91

2023

Reservoir Computing with Charge-Trap Memory Based on a MoS2 Channel for Neuromorphic Engineering

ADVANCED MATERIALS

Farronato, Matteo; Mannocci, Piergiulio; Melegari, Margherita; Ricci, Saverio; Compagnoni, Christian Monzio; Ielmini, Daniele

被引用次数: 39

2023

Threshold Switching by Bipolar Avalanche Multiplication in Ovonic Chalcogenide Glasses

ADVANCED ELECTRONIC MATERIALS

Fantini, Paolo; Polino, Nicola; Ghetti, Andrea; Ielmini, Daniele

被引用次数: 5

2023

A Multilayer Neural Accelerator With Binary Activations Based on Phase-Change Memory

IEEE TRANSACTIONS ON ELECTRON DEVICES

Bertuletti, Marco; Noz-Martin, Irene Mu; Bianchi, Stefano; Bonfanti, Andrea G.; Ielmini, Daniele

被引用次数: 7

2023

Process-Voltage-Temperature Variations Assessment in Energy-Aware Resistive RAM-Based FPGAs

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Rizzi, Tommaso; Baroni, Andrea; Glukhov, Artem; Bertozzi, Davide; Wenger, Christian; Ielmini, Daniele; Zambelli, Cristian

被引用次数: 1

2023

Tunable synaptic working memory with volatile memristive devices

NEUROMORPHIC COMPUTING AND ENGINEERING

Ricci, Saverio; Kappel, David; Tetzlaff, Christian; Ielmini, Daniele; Covi, Erika

被引用次数: 2

2023

A Generalized Block-Matrix Circuit for Closed-Loop Analog In-Memory Computing

IEEE JOURNAL ON EXPLORATORY SOLID-STATE COMPUTATIONAL DEVICES AND CIRCUITS

Mannocci, Piergiulio; Ielmini, Daniele

NimbleAI: Towards Neuromorphic Sensing-Processing 3D-integrated Chips

2023 DESIGN, AUTOMATION & TEST IN EUROPE CONFERENCE & EXHIBITION, DATE

Iturbe, Xabier; Abderrahmane, Nassim; Abella, Jaume; Alcaide, Sergi; Beyne, Eric; Charles, Henri-Pierre; Charpin-Nicolle, Christelle; Chittka, Lars; Davila, Angelica; Erdmann, Arne; Estrada, Carles; Fernandez, Ander; Fontanelli, Anna; Flich, Jose; Furano, Gianluca; Gloriani, Alejandro Hernan; Isusquiza, Erik; Grosu, Radu; Hernandez, Carles; Ielmini, Daniele; Jackson, David; Kooli, Maha; Lepri, Nicola; Linares-Barranco, Bernabe; Lachese, Jean-Loup; Laurent, Eric; Lindwer, Menno; Linsenmaier, Frank; Lujan, Mikel; Masarik, Karel; Mentens, Nele; Moreira, Orlando; Nawghane, Chinmay; Peres, Luca; Noel, Jean-Philippe; Pourtaherian, Arash; Posch, Christoph; Priller, Peter; Prikryl, Zdenek; Resch, Felix; Rhodes, Oliver; Stefanov, Todor; Storrington, Moritz; Taliercio, Michele; Tornero, Rafael; de Burgwal, Marcel van; van der Plas, Geert; Vianello, Elisa; Zaykov, Pavel

被引用次数: 1

2023

Invited Tutorial: Analog Matrix Computing With Crosspoint Resistive Memory Arrays

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Sun, Zhong; Ielmini, Daniele

被引用次数: 7

2022

Editorial: Novel materials, devices and solutions for brain-inspired sensing and computing

FRONTIERS IN NEUROSCIENCE

Monzio Compagnoni, Christian; Yu, Shimeng; Zhao, Weisheng; Ielmini, Daniele

被引用次数: 0

2022

In materia reservoir computing with a fully memristive architecture based on self-organizing nanowire networks

NATURE MATERIALS

Milano, Gianluca; Pedretti, Giacomo; Montano, Kevin; Ricci, Saverio; Hashemkhani, Shahin; Boarino, Luca; Ielmini, Daniele; Ricciardi, Carlo

被引用次数: 248

2022

Three-Terminal Ovonic Threshold Switch (3T-OTS) with Tunable Threshold Voltage for Versatile Artificial Sensory Neurons

NANO LETTERS

Lee, Hyejin; Cho, Seong Won; Kim, Seon Jeong; Lee, Jaesang; Kim, Keun Su; Kim, Inho; Park, Jong-Keuk; Kwak, Joon Young; Kim, Jaewook; Park, Jongkil; Jeong, YeonJoo; Hwang, Gyu Weon; Lee, Kyeong-Seok; Ielmini, Daniele; Lee, Suyoun

被引用次数: 15

2022

Memtransistor Devices Based on MoS₂ Multilayers with Volatile Switching due to Ag Cation Migration

ADVANCED ELECTRONIC MATERIALS

Farronato, Matteo; Melegari, Margherita; Ricci, Saverio; Hashemkhani, Shahin; Bricalli, Alessandro; Ielmini, Daniele

被引用次数: 24

2022

Forming-Free Resistive Switching Memory Crosspoint Arrays for In-Memory Machine Learning

ADVANCED INTELLIGENT SYSTEMS

Ricci, Saverio; Mannocci, Piergiulio; Farronato, Matteo; Hashemkhani, Shahin; Ielmini, Daniele

被引用次数: 13

2022

An Analogue In-Memory Ridge Regression Circuit With Application to Massive MIMO Acceleration

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Mannocci, Piergiulio; Melacarne, Enrico; Ielmini, Daniele

被引用次数: 7

2022

Low Conductance State Drift Characterization and Mitigation in Resistive Switching Memories (RRAM) for Artificial Neural Networks

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Baroni, Andrea; Glukhov, Artem; Perez, Eduardo; Wenger, Christian; Ielmini, Daniele; Olivo, Piero; Zambelli, Cristian

被引用次数: 11

2022

An energy-efficient in-memory computing architecture for survival data analysis based on resistive switching memories

FRONTIERS IN NEUROSCIENCE

Baroni, Andrea; Glukhov, Artem; Perez, Eduardo; Wenger, Christian; Calore, Enrico; Schifano, Sebastiano Fabio; Olivo, Piero; Ielmini, Daniele; Zambelli, Cristian

被引用次数: 5

2022

A CMOS-memristor hybrid system for implementing stochastic binary spike timing-dependent plasticity

PHILOSOPHICAL TRANSACTIONS OF THE ROYAL SOCIETY A-MATHEMATICAL PHYSICAL AND ENGINEERING SCIENCES

Ahmadi-Farsani, Javad; Ricci, Saverio; Hashemkhani, Shahin; Ielmini, Daniele; Linares-Barranco, Bernabe; Serrano-Gotarredona, Teresa

被引用次数: 9

2022

Ionic-electronic halide perovskite memdiodes enabling neuromorphic computing with a second-order complexity

SCIENCE ADVANCES

Abraham, Rohit; Kovalenko, Maksym V.; Ielmini, Daniele; Milozzi, Alessandro; Tsarev, Sergey; Bronnimann, Rolf; Boehme, Simon C.; Wu, Erfu; Shorubalko, Ivan

被引用次数: 15

2022

2022 roadmap on neuromorphic computing and engineering

NEUROMORPHIC COMPUTING AND ENGINEERING

Christensen, Dennis, V; Dittmann, Regina; Linares-Barranco, Bernabe; Sebastian, Abu; Le Gallo, Manuel; Redaelli, Andrea; Slesazeck, Stefan; Mikolajick, Thomas; Spiga, Sabina; Menzel, Stephan; Valov, Ilia; Milano, Gianluca; Ricciardi, Carlo; Liang, Shi-Jun; Miao, Feng; Lanza, Mario; Quill, Tyler J.; Keene, Scott T.; Salleo, Alberto; Grollier, Julie; Markovic, Danijela; Mizrahi, Alice; Yao, Peng; Yang, J. Joshua; Indiveri,

Giacomo; Strachan, John Paul; Datta, Suman; Vianello, Elisa; Valentian, Alexandre; Feldmann, Johannes; Li, Xuan; Pernice, Wolfram H. P.; Bhaskaran, Harish; Furber, Steve; Neftci, Emre; Scherr, Franz; Maass, Wolfgang; Ramaswamy, Srikanth; Tapsen, Jonathan; Panda, Priyadarshini; Kim, Youngeun; Tanaka, Gouhei; Thorpe, Simon; Bartolozzi, Chiara; Cleland, Thomas A.; Posch, Christoph; Liu, Shihchii; Panuccio, Gabriella; Mahmud, Mufti; Mazumder, Arnab Neelim; Hosseini, Morteza; Mohsenin, Tinoosh; Donati, Elisa; Tolu, Silvia; Galeazzi, Roberto; Christensen, Martin Ejsing; Holm, Sune; Ielmini, Daniele; Pryds, N.

被引用次数: 369

2022

Exploring Process-Voltage-Temperature Variations Impact on 4T1R Multiplexers for Energy-aware Resistive RAM-based FPGAs

2022 IEEE INTERNATIONAL INTEGRATED RELIABILITY WORKSHOP, IIRW

Rizzi, Tommaso; Baroni, Andrea; Glukhov, Artem; Bertozzi, Davide; Wenger, Christian; Ielmini, Daniele; Zambelli, Cristian

被引用次数: 1

2022

A Hybrid Memristor/CMOS SNN for Implementing One-Shot Winner-Takes-All Training

2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)

Ahmadi-Farsani, Javad; Ricci, Saverio; Hashemkhani, Shahin; Ielmini, Daniele; Linares-Barranco, Bernabe; Serrano-Gotarredona, Teresa

被引用次数: 0

2022

Experimental verification and benchmark of in-memory principal component analysis by crosspoint arrays of resistive switching memory

2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)

Mannocci, Piergiulio; Baroni, Andrea; Melacarne, Enrico; Zambelli, Cristian; Olivo, Piero; Perez, Eduardo; Wenger, Christian; Ielmini, Daniele

被引用次数: 0

2022

Interaction between forming pulse and integration process flow in ePCM

PRIME 2022: 17TH INTERNATIONAL CONFERENCE ON PHD RESEARCH IN MICROELECTRONICS AND ELECTRONICS

Baldo, Matteo; Petroni, Elisa; Laurin, Luca; Samanni, Giulia; Melnic, Octavian; Ielmini, Daniele; Redaelli, Andrea

被引用次数: 1

2022

Statistical model of program/verify algorithms in resistive-switching memories for in-memory neural network accelerators

2022 IEEE INTERNATIONAL RELIABILITY PHYSICS SYMPOSIUM (IRPS)

Glukhov, Artem; Milo, Valerio; Baroni, Andrea; Lepri, Nicola; Zambelli, Cristian; Olivo, Piero; Perez, Eduardo; Wenger, Christian; Ielmini, Daniele

被引用次数: 0

2022

Statistical Model of Program/Verify Algorithms in Resistive-Switching Memories for In-Memory Neural Network Accelerators

2022 IEEE INTERNATIONAL RELIABILITY PHYSICS SYMPOSIUM (IRPS)

Glukhov, Artem; Lepri, Nicola; Ielmini, Daniele; Milo, Valerio; Baroni, Andrea; Perez, Eduardo; Zambelli, Cristian; Olivo, Piero; Wenger, Christian

被引用次数: 3

2022

Decision Making by a Neuromorphic Network of Volatile Resistive Switching Memories

2022 29TH IEEE INTERNATIONAL CONFERENCE ON ELECTRONICS, CIRCUITS AND SYSTEMS (IEEE ICECS 2022)

Ricci, Saverio; Kappel, David; Tetzlaff, Christian; Ielmini, Daniele; Covi, Erika

被引用次数: 1

2022

Characterization of reset state through energy activation study in Ge-GST based ePCM

ESSDERC 2022 - IEEE 52ND EUROPEAN SOLID-STATE DEVICE RESEARCH CONFERENCE (ESSDERC)

Baldo, Matteo; Turconi, Lorenzo; Motta, Alessandro; Petroni, Elisa; Laurin, Luca; Ielmini, Daniele; Redaelli, Andrea

被引用次数: 4

2022

End-to-end modeling of variability-aware neural networks based on resistive-switching memory arrays

PROCEEDINGS OF THE 2022 IFIP/IEEE 30TH INTERNATIONAL CONFERENCE ON VERY LARGE SCALE INTEGRATION (VLSI-SOC)

Glukhov, Artem; Lepri, Nicola; Milo, Valerio; Baroni, Andrea; Zambelli, Cristian; Olivo, Piero; Perez, Eduardo; Wenger, Christian; Ielmini, Daniele

被引用次数: 1

2022

Standards for the Characterization of Endurance in Resistive Switching Devices

ACS NANO

Lanza, Mario; Waser, Rainer; Ielmini, Daniele; Yang, J. Joshua; Goux, Ludovic; Sune, Jordi; Kenyon, Anthony Joseph; Mehonic, Adnan; Spiga, Sabina; Rana, Vikas; Wiefels, Stefan; Menzel, Stephan; Valov, Ilia; Villena, Marco A.; Miranda, Enrique; Jing, Xu; Campabadal, Francesca; Gonzalez, Mireia B.; Aguirre, Fernando; Palumbo, Felix; Zhu, Kaichen; Roldan, Juan Bautista; Puglisi, Francesco Maria; Larcher, Luca; Hou, Tuo-Hung; Prodromakis, Themis; Yang, Yuchao; Huang, Peng; Wan, Tianqing; Chai, Yang; Pey, Kin Leong; Raghavan, Nagarajan; Duenas, Salvador; Wang, Tao; Xia, Qiangfei; Pazos, Sebastian

被引用次数: 137

2021

Optimization Schemes for In-Memory Linear Regression Circuit With Memristor Arrays

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Wang, Shiqing; Sun, Zhong; Liu, Yuheng; Bao, Shengyu; Cai, Yimao; Ielmini, Daniele; Huang, Ru

被引用次数: 8

2021

Neuromorphic Motion Detection and Orientation Selectivity by Volatile Resistive Switching Memories

ADVANCED INTELLIGENT SYSTEMS

Wang, Wei; Covi, Erika; Milozzi, Alessandro; Farronato, Matteo; Ricci, Saverio; Sbandati, Caterina; Pedretti, Giacomo; Ielmini, Daniele

被引用次数: 50

2021

A Universal, Analog, In-Memory Computing Primitive for Linear Algebra Using Memristors

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Mannocci, Piergiulio; Pedretti, Giacomo; Giannone, Elisabetta; Melacarne, Enrico; Sun, Zhong; Ielmini, Daniele

被引用次数: 13

2021

A Drift-Resilient Hardware Implementation of Neural Accelerators Based on Phase Change Memory Devices

IEEE TRANSACTIONS ON ELECTRON DEVICES

Munoz-Martin, Irene; Bianchi, Stefano; Melnic, Octavian; Giovanni Bonfanti, Andrea; Ielmini, Daniele

被引用次数: 12

2021

Switching Dynamics of Ag-Based Filamentary Volatile Resistive Switching Devices-Part I: Experimental Characterization

IEEE TRANSACTIONS ON ELECTRON DEVICES

Covi, Erika; Wang, Wei; Lin, Yu-Hsuan; Farronato, Matteo; Ambrosi, Elia; Ielmini, Daniele

被引用次数: 37

2021

Switching Dynamics of Ag-Based Filamentary Volatile Resistive Switching Devices-Part II: Mechanism and Modeling

IEEE TRANSACTIONS ON ELECTRON DEVICES

Wang, Wei; Covi, Erika; Lin, Yu-Hsuan; Ambrosi, Elia; Milozzi, Alessandro; Sbandati, Caterina; Farronato, Matteo; Ielmini, Daniele

被引用次数: 36

2021

Redundancy and Analog Slicing for Precise In-Memory Machine Learning-Part I: Programming Techniques

IEEE TRANSACTIONS ON ELECTRON DEVICES

Pedretti, Giacomo; Mannocci, Piergiulio; Li, Can; Sun, Zhong; Strachan, John Paul; Ielmini, Daniele

被引用次数: 16

2021

Redundancy and Analog Slicing for Precise In-Memory Machine Learning-Part II: Applications and Benchmark

IEEE TRANSACTIONS ON ELECTRON DEVICES

Pedretti, Giacomo; Mannocci, Piergiulio; Li, Can; Sun, Zhong; Strachan, John Paul; Ielmini, Daniele

被引用次数: 5

2021

Accurate Program/Verify Schemes of Resistive Switching Memory (RRAM) for In-Memory Neural Network Circuits

IEEE TRANSACTIONS ON ELECTRON DEVICES

Milo, Valerio; Glukhov, Artem; Perez, Eduardo; Zambelli, Cristian; Lepri, Nicola; Mahadevaiah, Mamathamba Kalishettyhalli; Quesada, Emilio Perez-Bosch; Olivo, Piero; Wenger, Christian; Ielmini, Daniele

被引用次数: 58

2021

In-Memory Computing with Resistive Memory Circuits: Status and Outlook

ELECTRONICS

Pedretti, Giacomo; Ielmini, Daniele

被引用次数: 44

2021

Empirical metal-oxide RRAM device endurance and retention model for deep learning simulations

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Lammie, Corey; Rahimi Azghadi, Mostafa; Ielmini, Daniele

被引用次数: 10

2021

A Brain-Inspired Homeostatic Neuron Based on Phase-Change Memories for Efficient Neuromorphic Computing

FRONTIERS IN NEUROSCIENCE

Munoz-Martin, Irene; Bianchi, Stefano; Hashemkhani, Shahin; Pedretti, Giacomo; Melnic, Octavian; Ielmini, Daniele

被引用次数: 8

2021

Combining Accuracy and Plasticity in Convolutional Neural Networks Based on Resistive Memory Arrays for Autonomous Learning

IEEE JOURNAL ON EXPLORATORY SOLID-STATE COMPUTATIONAL DEVICES AND CIRCUITS

Bianchi, Stefano; Munoz-Martin, Irene; Covi, Erika; Bricalli, Alessandro; Piccoloni, Giuseppe; Regev, Amir; Molas, Gabriel; Nodin, Jean Francois; Andrieu, Francois; Ielmini, Daniele

被引用次数: 2

2021

Tackling the Low Conductance State Drift through Incremental Reset and Verify in RRAM arrays

2021 IEEE INTERNATIONAL INTEGRATED RELIABILITY WORKSHOP (IIRW)

Baroni, Andrea; Zambelli, Cristian; Olivo, Piero; Perez, Eduardo; Wenger, Christian; Ielmini, Daniele

被引用次数: 3

2021

Toward a generalized Bienenstock-Cooper-Munro rule for spatiotemporal learning via triplet-STDP in memristive devices

NATURE COMMUNICATIONS

Wang, Zhongqiang; Zeng, Tao; Ren, Yanyun; Lin, Ya; Xu, Haiyang; Zhao, Xiaoning; Liu, Yichun; Ielmini, Daniele

被引用次数: 144

2020

Memristive and CMOS Devices for Neuromorphic Computing

MATERIALS

Milo, Valerio; Malavena, Gerardo; Compagnoni, Christian Monzio; Ielmini, Daniele

被引用次数: 95

2020

Emerging neuromorphic devices

NANOTECHNOLOGY

Ielmini, Daniele; Ambrogio, Stefano

被引用次数: 214

2020

In-Memory Eigenvector Computation in Time $O(1)$

ADVANCED INTELLIGENT SYSTEMS

Sun, Zhong; Pedretti, Giacomo; Ambrosi, Elia; Bricalli, Alessandro; Ielmini, Daniele

被引用次数: 16

2020

Brain-Inspired Structural Plasticity through Reweighting and Rewiring in Multi-Terminal Self-Organizing Memristive Nanowire Networks

ADVANCED INTELLIGENT SYSTEMS

Milano, Gianluca; Pedretti, Giacomo; Fretto, Matteo; Boarino, Luca; Benfenati, Fabio; Ielmini, Daniele; Valov, Ilia; Ricciardi, Carlo

被引用次数: 85

2020

Device and Circuit Architectures for In-Memory Computing

ADVANCED INTELLIGENT SYSTEMS

Ielmini, Daniele; Pedretti, Giacomo

被引用次数: 106

2020

Reliability of Logic-in-Memory Circuits in Resistive Memory Arrays

IEEE TRANSACTIONS ON ELECTRON DEVICES

Zanotti, Tommaso; Zambelli, Cristian; Puglisi, Francesco Maria; Milo, Valerio; Perez, Eduardo; Mahadevaiah, Mamathamba K.; Ossorio, Oscar G.; Wenger, Christian; Pavan, Paolo; Olivo, Piero; Ielmini, Daniele

被引用次数: 10

2020

A Compact Model for Stochastic Spike-Timing-Dependent Plasticity (STDP) Based on Resistive Switching Memory (RRAM) Synapses

IEEE TRANSACTIONS ON ELECTRON DEVICES

Bianchi, Stefano; Pedretti, Giacomo; Munoz-Martin, Irene; Calderoni, Alessandro; Ramaswamy, Nirmal; Ambrogio, Stefano; Ielmini, Daniele

被引用次数: 12

2020

Time Complexity of In-Memory Solution of Linear Systems

IEEE TRANSACTIONS ON ELECTRON DEVICES

Sun, Zhong; Pedretti, Giacomo; Mannocci, Piergiulio; Ambrosi, Elia; Bricalli, Alessandro; Ielmini, Daniele

被引用次数: 24

2020

In-Memory PageRank Accelerator With a Cross-Point Array of Resistive Memories

IEEE TRANSACTIONS ON ELECTRON DEVICES

Sun, Zhong; Ambrosi, Elia; Pedretti, Giacomo; Bricalli, Alessandro; Ielmini, Daniele

被引用次数: 36

2020

Bio-Inspired Techniques in a Fully Digital Approach for Lifelong Learning

FRONTIERS IN NEUROSCIENCE

Bianchi, Stefano; Munoz-Martin, Irene; Ielmini, Daniele

被引用次数: 7

2020

Integration and Co-design of Memristive Devices and Algorithms for Artificial Intelligence

ISCIENCE

Wang, Wei; Song, Wenhao; Yao, Peng; Li, Yang; Van Nostrand, Joseph; Qiu, Qinru; Ielmini, Daniele; Yang, J. Joshua

被引用次数: 59

2020

One-step regression and classification with cross-point resistive memory arrays

SCIENCE ADVANCES

Sun, Zhong; Pedretti, Giacomo; Bricalli, Alessandro; Ielmini, Daniele

被引用次数: 71

2020

Recommended Methods to Study Resistive Switching Devices

ADVANCED ELECTRONIC MATERIALS

Lanza, Mario; Wong, H-S Philip; Pop, Eric; Ielmini, Daniele; Strukov, Dimitri; Regan, Brian C.; Larcher, Luca; Villena, Marco A.; Yang, J. Joshua; Goux, Ludovic; Belmonte, Attilio; Yang, Yuchao; Puglisi, Francesco M.; Kang, Jinfeng; Magyari-Kope, Blanka; Yalon, Eilam; Kenyon, Anthony; Buckwell, Mark; Mehonic, Adnan; Shluger, Alexander; Li, Haitong; Hou, Tuo-Hung; Hudec, Boris; Akinwande, Deji; Ge, Ruijing; Ambrogio, Stefano; Roldan, Juan B.; Miranda, Enrique; Sune, Jordi; Pey, Kin Leong; Wu, Xing; Raghavan, Nagarajan; Wu, Ernest; Lu, Wei D.; Navarro, Gabriele; Zhang, Weidong; Wu, Huaqiang; Li, Runwei; Holleitner, Alexander; Wurstbauer, Ursula; Lemme, Max C.; Liu, Ming; Long, Shibing; Liu, Qi; Lv, Hangbing; Padovani, Andrea; Pavan, Paolo; Valov, Ilia; Jing, Xu; Han, Tingting; Zhu, Kaichen; Chen, Shaochuan; Hui, Fei; Shi, Yuanyuan

被引用次数: 463

2019

Stochastic Memory Devices for Security and Computing

ADVANCED ELECTRONIC MATERIALS

Carboni, Roberto; Ielmini, Daniele

被引用次数: 105

2019

Impact of oxide and electrode materials on the switching characteristics of oxide ReRAM devices

FARADAY DISCUSSIONS

Ambrosi, Elia; Bricalli, Alessandro; Laudato, Mario; Ielmini, Daniele

被引用次数: 82

2019

A Physics-Based Compact Model of Stochastic Switching in Spin-Transfer Torque Magnetic Memory

IEEE TRANSACTIONS ON ELECTRON DEVICES

Carboni, Roberto; Vernocchi, Elena; Siddik, Manzar; Harms, Jon; Lyle, Andy; Sandhu, Gurtej; Ielmini, Daniele

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2019

Volatile Resistive Switching Memory Based on Ag Ion Drift/Diffusion Part I: Numerical Modeling

IEEE TRANSACTIONS ON ELECTRON DEVICES

Wang, Wei; Laudato, Mario; Ambrosi, Elia; Bricalli, Alessandro; Covi, Erika; Lin, Yu-Hsuan; Ielmini, Daniele

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2019

Volatile Resistive Switching Memory Based on Ag Ion Drift/Diffusion-Part II: Compact Modeling

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2019

Solving matrix equations in one step with cross-point resistive arrays

PROCEEDINGS OF THE NATIONAL ACADEMY OF SCIENCES OF THE UNITED STATES OF AMERICA

Sun, Zhong; Pedretti, Giacomo; Ambrosi, Elia; Bricalli, Alessandro; Wang, Wei; Ielmini, Daniele

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2019

Surface diffusion-limited lifetime of silver and copper nanofilaments in resistive switching devices

NATURE COMMUNICATIONS

Wang, Wei; Wang, Ming; Ambrosi, Elia; Bricalli, Alessandro; Laudato, Mario; Sun, Zhong; Chen, Xiaodong; Ielmini, Daniele

被引用次数: 405

2019

Silicon Oxide (SiO_x): A Promising Material for Resistance Switching?

ADVANCED MATERIALS

Mehonic, Adnan; Shluger, Alexander L.; Gao, David; Valov, Ilia; Miranda, Enrique; Ielmini, Daniele; Bricalli, Alessandro; Ambrosi, Elia; Li, Can; Yang, J. Joshua; Xia, Qiangfei; Kenyon, Anthony J.

被引用次数: 160

2018

Logic Computing with Stateful Neural Networks of Resistive Switches

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2018

Enhancing the Matrix Addressing of Flexible Sensory Arrays by a Highly Nonlinear Threshold Switch

ADVANCED MATERIALS

Wang, Ming; Wang, Wei; Leow, Wan Ru; Wan, Changjin; Chen, Geng; Zeng, Yi; Yu, Jiancan; Liu, Yaqing; Cai, Pingqiang; Wang, Hong; Ielmini, Daniele; Chen, Xiaodong

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2018

In-memory computing with resistive switching devices

NATURE ELECTRONICS

Ielmini, Daniele; Wong, H. -S. Philip

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2018

Random Number Generation by Differential Read of Stochastic Switching in Spin-Transfer Torque Memory

IEEE ELECTRON DEVICE LETTERS

Carboni, Roberto; Chen, Wei; Siddik, Manzar; Harms, Jon; Lyle, Andy; Kula, Witold; Sandhu, Gurtej; Ielmini, Daniele

被引用次数: 14

2018

Stochastic Learning in Neuromorphic Hardware via Spike Timing Dependent Plasticity With RRAM Synapses

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Pedretti, Giacomo; Milo, Valerio; Ambrogio, Stefano; Carboni, Roberto; Bianchi, Stefano; Calderoni, Alessandro; Ramaswamy, Nirmal; Spinelli, Alessandro S.; Ielmini, Daniele

Modeling of Breakdown-Limited Endurance in Spin-Transfer Torque Magnetic Memory Under Pulsed Cycling Regime

IEEE TRANSACTIONS ON ELECTRON DEVICES

Carboni, Roberto; Ambrogio, Stefano; Chen, Wei; Siddik, Manzar; Harms, Jon; Lyle, Andy; Kula, Witold; Sandhu, Gurtej; Ielmini, Daniele

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2018

Resistive Switching Device Technology Based on Silicon Oxide for Improved ON-OFF Ratio-Part I: Memory Devices

IEEE TRANSACTIONS ON ELECTRON DEVICES

Bricalli, Alessandro; Ambrosi, Elia; Laudato, Mario; Maestro, Marcos; Rodriguez, Rosana; Ielmini, Daniele

被引用次数: 56

2018

Brain-inspired computing with resistive switching memory (RRAM): Devices, synapses and neural networks

MICROELECTRONIC ENGINEERING

Ielmini, Daniele

被引用次数: 240

2018

Learning of spatiotemporal patterns in a spiking neural network with resistive switching synapses

SCIENCE ADVANCES

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2018

Editorial for the JECR special issue on resistive switching: Oxide materials, mechanisms, and devices

JOURNAL OF ELECTROCERAMICS

Rupp, Jennifer L. M.; Valov, Ilia; Ielmini, Daniele

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2017

Neuromorphic Learning and Recognition With One-Transistor-One-Resistor Synapses and Bistable Metal Oxide RRAM

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2016

Resistive switching memories based on metal oxides: mechanisms, reliability and scaling

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Ielmini, Daniele

被引用次数: 698

2016

Unsupervised Learning by Spike Timing Dependent Plasticity in Phase Change Memory (PCM) Synapses

FRONTIERS IN NEUROSCIENCE

Ambrogio, Stefano; Ciochini, Nicola; Laudato, Mario; Milo, Valerio; Pirovano, Agostino; Fantini, Paolo; Ielmini, Daniele

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2016

True Random Number Generation by Variability of Resistive Switching in Oxide-Based Devices

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Balatti, Simone; Ambrogio, Stefano; Wang, Zhongqiang; Ielmini, Daniele

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2015

Normally-off Logic Based on Resistive Switches-Part I: Logic Gates

IEEE TRANSACTIONS ON ELECTRON DEVICES

Balatti, Simone; Ambrogio, Stefano; Ielmini, Daniele

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2015

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IEEE TRANSACTIONS ON ELECTRON DEVICES

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2015

A 2-transistor/1-resistor artificial synapse capable of communication and stochastic learning in neuromorphic systems

FRONTIERS IN NEUROSCIENCE

Wang, Zhongqiang; Ambrogio, Stefano; Balatti, Simone; Ielmini, Daniele

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2015

Impact of the Mechanical Stress on Switching Characteristics of Electrochemical Resistive Memory

ADVANCED MATERIALS

Ambrogio, Stefano; Balatti, Simone; Choi, Seol; Ielmini, Daniele

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2014

Statistical Fluctuations in HfOx Resistive-Switching Memory: Part I - Set/Reset Variability

IEEE TRANSACTIONS ON ELECTRON DEVICES

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Statistical Fluctuations in HfOx Resistive-Switching Memory: Part II-Random Telegraph Noise

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Analytical Modeling of Oxide-Based Bipolar Resistive Memories and Complementary Resistive Switches

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Ambrogio, Stefano; Balatti, Simone; Gilmer, David C.; Ielmini, Daniele

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2014

Modeling Resistance Instabilities of Set and Reset States in Phase Change Memory With Ge-Rich GeSbTe

IEEE TRANSACTIONS ON ELECTRON DEVICES

Ciocchini, Nicola; Palumbo, Elisabetta; Borghi, Massimo; Zuliani, Paola; Annunziata, Roberto; Ielmini, Daniele

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2014

Filament Evolution during Set and Reset Transitions in Oxide Resistive Switching Memory

JAPANESE JOURNAL OF APPLIED PHYSICS

Ielmini, Daniele; Balatti, Simone; Larentis, Stefano

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Self-Aligned Nanotube-Nanowire Phase Change Memory

NANO LETTERS

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Set Variability and Failure Induced by Complementary Switching in Bipolar RRAM

IEEE ELECTRON DEVICE LETTERS

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Energy Landscape Model of Conduction and Phase Transition in Phase Change Memories

IEEE TRANSACTIONS ON ELECTRON DEVICES

Rizzi, Maurizio; Ferro, Massimo; Fantini, Paolo; Ielmini, Daniele

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2013

Evidence for Non-Arrhenius Kinetics of Crystallization in Phase Change Memory Devices

IEEE TRANSACTIONS ON ELECTRON DEVICES

Ciocchini, Nicola; Cassinero, Marco; Fugazza, Davide; Ielmini, Daniele

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Complementary Switching in Oxide-Based Bipolar Resistive-Switching Random Memory

IEEE TRANSACTIONS ON ELECTRON DEVICES

Nardi, Federico; Balatti, Simone; Larentis, Stefano; Gilmer, David C.; Ielmini, Daniele

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Hot-carrier trap-limited transport in switching chalcogenides

JOURNAL OF APPLIED PHYSICS

Piccinini, Enrico; Cappelli, Andrea; Buscemi, Fabrizio; Brunetti, Rossella; Ielmini, Daniele; Rudan, Massimo; Jacoboni, Carlo

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Analysis and modeling of resistive switching statistics

JOURNAL OF APPLIED PHYSICS

Long, Shibing; Cagli, Carlo; Ielmini, Daniele; Liu, Ming; Sune, Jordi

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Phase change materials

MRS BULLETIN

Raoux, Simone; Ielmini, Daniele; Wuttig, Matthias; Karpov, Ilya

被引用次数: 81

2012

Size-Dependent Drift of Resistance Due to Surface Defect Relaxation in Conductive-Bridge Memory

IEEE ELECTRON DEVICE LETTERS

Choi, Seol; Balatti, Simone; Nardi, Federico; Ielmini, Daniele

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Modeling of Threshold-Voltage Drift in Phase-Change Memory (PCM) Devices

IEEE TRANSACTIONS ON ELECTRON DEVICES

Ciocchini, Nicola; Cassinerio, Marco; Fugazza, Davide; Ielmini, Daniele

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Resistive Switching by Voltage-Driven Ion Migration in Bipolar RRAM-Part I: Experimental Study

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Unified physical modeling of reliability mechanisms and scaling perspective of phase change memory

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Ielmini, Daniele

Impact of Ge-Sb-Te compound engineering on the set operation performance in phase-change memories

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Boniardi, Mattia; Ielmini, Daniele; Tortorelli, Innocenzo; Redaelli, Andrea; Pirovano, Agostino; Allegra, Mario; Magistretti, Michele; Bresolin, Camillo; Erbetta, Davide; Modelli, Alberto; Varesi, Enrico; Pellizzer, Fabio; Lacaïta, Andrea L.; Bez, Roberto

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2011

Phase change materials in non-volatile storage

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Ielmini, Daniele; Lacaïta, Andrea L.

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2011

Resistive-Switching Crossbar Memory Based on Ni-NiO Core-Shell Nanowires

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Thermochemical resistive switching: materials, mechanisms, and scaling projections

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Ielmini, Daniele; Bruchhaus, Rainer; Waser, Rainer

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Reset Statistics of NiO-Based Resistive Switching Memories

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Long, Shibing; Cagli, Carlo; Ielmini, Daniele; Liu, Ming; Sune, Jordi

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Reset Instability in Pulsed-Operated Unipolar Resistive-Switching Random Access Memory Devices

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Nardi, Federico; Cagli, Carlo; Spiga, Sabina; Ielmini, Daniele

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Modeling the Universal Set/Reset Characteristics of Bipolar RRAM by Field- and Temperature-Driven Filament Growth

IEEE TRANSACTIONS ON ELECTRON DEVICES

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Resistance-dependent amplitude of random telegraph-signal noise in resistive switching memories

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Phase Change Materials and Their Application to Nonvolatile Memories

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Raoux, Simone; Welnic, Wojciech; Ielmini, Daniele

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A Unified Hopping Model for Subthreshold Current of Phase-Change Memories in Amorphous State

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Calderoni, Alessandro; Ferro, Massimo; Ielmini, Daniele; Fantini, Paolo

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IEEE ELECTRON DEVICE LETTERS

Ielmini, Daniele

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2010

Incomplete Filament Crystallization During Set Operation in PCM Cells

IEEE ELECTRON DEVICE LETTERS

Mantegazza, Davide; Ielmini, Daniele; Pirovano, Agostino; Lacaita, Andrea L.

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Size-Dependent Retention Time in NiO-Based Resistive-Switching Memories

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Ielmini, Daniele; Nardi, Federico; Cagli, Carlo; Lacaita, Andrea L.

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Lavizzari, Simone; Ielmini, Daniele; Lacaita, Andrea L.

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Statistics of Resistance Drift Due to Structural Relaxation in Phase-Change Memory Arrays

IEEE TRANSACTIONS ON ELECTRON DEVICES

Boniardi, Mattia; Ielmini, Daniele; Lavizzari, Simone; Lacaita, Andrea L.; Redaelli, Andrea; Pirovano, Agostino

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Threshold-Switching Delay Controlled by $1/f$ Current Fluctuations in Phase-Change Memory Devices

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lavizzari, Simone; Sharma, Deepak; Ielmini, Daniele

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2010

Common signature of many-body thermal excitation in structural relaxation and crystallization of chalcogenide glasses

APPLIED PHYSICS LETTERS

Ielmini, Daniele; Boniardi, Mattia

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Impact of Electrode Materials on Resistive-Switching Memory Programming

IEEE ELECTRON DEVICE LETTERS

Russo, Ugo; Cagli, Carlo; Spiga, Sabina; Cianci, Elena; Ielmini, Daniele

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Delay Correction for Accurate Extraction of Time Exponent and Activation Energy of NBTI

IEEE ELECTRON DEVICE LETTERS

Ielmini, Daniele; Gattel, Francesco

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Voltage-Driven On-Off Transition and Tradeoff With Program and Erase Current in Programmable Metallization Cell (PMC) Memory

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Kamalanathan, Deepak; Russo, Ugo; Ielmini, Daniele; Kozicki, Michael N.

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2009

A New NBTI Model Based on Hole Trapping and Structural Relaxation in MOS Dielectrics

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Ielmini, Daniele; Manigrasso, Mariaflavia; Gattel, Francesco; Valentini, Maria Grazia

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Modeling of Set/Reset Operations in NiO-Based Resistive-Switching Memory Devices

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Cagli, Carlo; Nardi, Federico; Ielmini, Daniele

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Study of Multilevel Programming in Programmable Metallization Cell (PMC) Memory

IEEE TRANSACTIONS ON ELECTRON DEVICES

Russo, Ugo; Kamalanathan, Deepak; Ielmini, Daniele; Lacaita, Andrea L.; Kozicki, Michael N.

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MICROELECTRONIC ENGINEERING

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Threshold switching mechanism by high-field energy gain in the hopping transport of chalcogenide glasses

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Ielmini, Daniele

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Voltage-controlled relaxation oscillations in phase-change memory devices

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Modeling of programming and read performance in phase-change memories - Part I: Cell optimization and scaling

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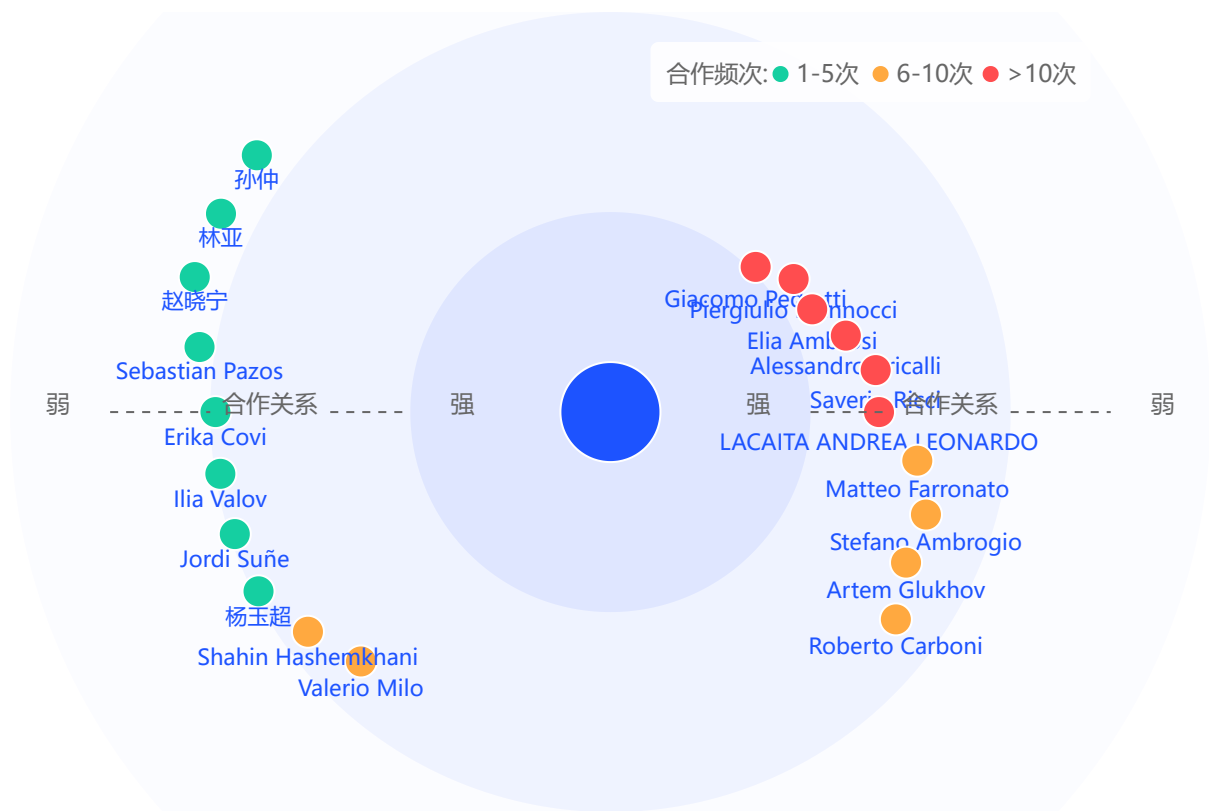
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学术合作



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米兰理工大学

合作次数: 14

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合作次数: 13

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